Now let us look at an important way of describing the relationship between two quantities using the notion of a function.

(i) Slope and Intercept form

Proportional linear functions can be written in the form $y = mx$, where m is the slope of the line. Non proportional linear functions can be written in the form

$$y = m\,x + b, \quad b \neq 0 \tag{6.5}$$

This is called the slope-intercept form of a straight line because m is the slope and b is the y-intercept.

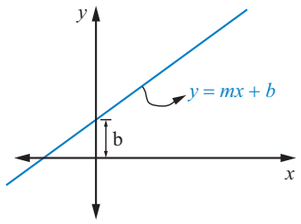


Figure 6.18

(1) when $b = 0$ and $m \neq 0$, the line passes through the origin and its equation is $y = mx$.
(2) when $b = 0$ and $m = 0$, the line coincides with the x-axis and its equation is $y = 0$.
(3) when $b \neq 0$ and $m = 0$, the line is parallel to the x-axis and its equation is $y = b$.

(ii) Point - Slope form:

Let m be the slope of the line and $A(x_1, y_1)$ be the given point on the line. Let $P(x, y)$ be any point other than A on the given line. Slope of the line joining $A(x_1, y_1)$ and $P(x, y)$ is given by $m = \dfrac{y - y_1}{x - x_1}$

$$\Rightarrow y - y_1 = m(x - x_1), \tag{6.6}$$

which is known as point-slope form.

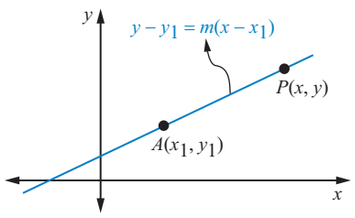


Figure 6.19

Since, the slope m is undefined for lines parallel to the y-axis, the point-slope form of the equation will not give the equation of a line through $A(x_1, y_1)$ parallel to the y-axis. However, this presents no difficulty, since for any such line the abscissa of any point on the line is x_1. Therefore, the equation of such a line is $x = x_1$.

(iii) Two Points form

If (x_1, y_1) and (x_2, y_2) are any two points on the line with $x_2 \neq x_1$ and $y_1 \neq y_2$, then the slope is $m = \dfrac{(y_2 - y_1)}{(x_2 - x_1)}$.

The equation using point- slope form, we get $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.

Rewriting the above equation, we get

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \tag{6.7}$$

This equation is called two points form.

Two points form can also be represented in terms of the determinant as

$$\begin{vmatrix} x - x_1 & y - y_1 \\ x_2 - x_1 & y_2 - y_1 \end{vmatrix} = 0.$$

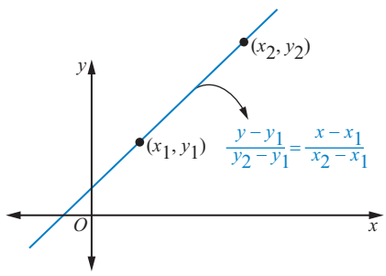


Figure 6.20